

HOW ACCURATE IS GOOD ENOUGH?

Approximating a Civil Engineering Function Using Infinite Series

An Integrated Python Exercise for Civil Engineering Students

Topics Covered:

- Geometric Series
 - Power Series
- Maclaurin Series
 - Taylor Series
- Error Analysis and Engineering Decision-Making

Learning Outcomes:

Students will understand how infinite series can approximate engineering functions, quantify approximation errors, and make engineering decisions based on accuracy requirements.

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ENGINEERING SCENARIO

A civil engineer is analyzing the geometry of a sloping structural or surveying system. The relationship between angle and vertical component is given by:

$$y = L \sin(\theta)$$

Where:

- $L = 20$ m (structural length)
- θ = angle in radians
- y = vertical component

Central Question: How accurately can a civil engineer approximate an engineering quantity using a finite number of terms of an infinite series?

I can approximate an engineering quantity quite accurately using only a few terms, as long as the angle is within a suitable range. In my results, 2 Maclaurin terms were enough to keep the error below 0.1% for angles from 1° to 30° .

PART 1: GEOMETRIC SERIES

Begin with the geometric series and its partial sum:

$$1 + x + x^2 + x^3 + \dots = \sum x^k$$

Task: Implement a Python function to calculate the partial sum without using the closed-form equation $1/(1-x)$.

```
def geometric_sum(x, N): """Calculate S_N = 1 + x + x^2 + ... + x^N""" total = 0.0 for k in range(N + 1): total += x ** k return total
```

Investigation: Compare partial sums with the exact value for $x = 0.5$, 0.8 , and 0.9 . How does the number of terms affect accuracy? Why does convergence speed differ?

Key Concepts: partial sum, convergence, approximation error, number of terms needed

How does the number of terms affect accuracy?

The more terms I use, the more accurate the answer becomes.

Why does convergence speed differ?

It happens faster when x is smaller because the terms become smaller more quickly.

PART 2: POWER SERIES

Generalize the geometric series into a power series:

$$P_N(x) = a_0 + a_1x + a_2x^2 + \dots + a_Nx^N$$

Task: Create a function that evaluates a power series at a given point.

```
def power_series(x, coefficients): """Evaluate P_N(x) with given coefficients""" result
= 0.0 for k, a k in enumerate(coefficients): result += a k * (x ** k) return result
```

Key Insight: A polynomial is a finite power series. An infinite power series can approximate transcendental functions like sin, cos, and exponentials.

PART 3: MACLAURIN SERIES FOR $\sin(\theta)$

The Maclaurin series is a power series centered at $\theta = 0$:

$$\sin(\theta) = \theta - \theta^3/3! + \theta^5/5! - \theta^7/7! + \dots$$

Task: Build the Maclaurin approximation explicitly using loops. Do NOT use `math.sin()` to generate the approximation (only for verification).

```
def sin_maclaurin(theta, N): """Approximate sin(theta) with N terms""" result = 0.0
for n in range(N): sign = (-1) ** n factorial = math.factorial(2*n + 1) result += sign *
(theta ** (2*n + 1)) / factorial return result
```

Investigation: For $\theta = 10^\circ$, calculate approximations using 1, 2, 3, and 4 terms. Compare each to the exact value. What is the error at each step?

As I added more terms, the approximation became closer to the exact value. With 2 terms, the error was already below 0.1%, so 2 terms were enough for this requirement.

PART 4: ENGINEERING INVESTIGATION

Engineering Problem: Calculate the vertical component $y = L \sin(\theta)$ for various angles using Maclaurin approximations.

Given Data:

- $L = 20 \text{ m}$
- Angles: $1^\circ, 2^\circ, 5^\circ, 10^\circ, 15^\circ, 20^\circ, 30^\circ$
- Use Maclaurin approximations with 1, 2, 3, and 4 terms

Task: Create a table for each number of terms showing:

- Angle (degrees)
- Exact value y
- Approximation
- Absolute error
- Percentage error

Analysis Questions:

1. How does the error change as the angle increases?
The error generally increases as the angle gets bigger.
2. How does the error change as you add more terms?
The error becomes smaller as I add more terms.
3. For small angles ($< 5^\circ$), how many terms are sufficient?
1 term is enough for small angles.
4. For larger angles ($> 20^\circ$), how many terms are needed?
2 terms are enough based on my results.

PART 5: TAYLOR SERIES

The Taylor series generalizes from Maclaurin by expanding around any point a :

$$f(\theta) = f(a) + f'(a)(\theta - a) + f''(a)/2!(\theta - a)^2 + \dots$$

Key Insight: Maclaurin is simply Taylor with $a = 0$. By choosing a different expansion point, you can improve accuracy in different regions.

Task: Construct a Taylor approximation for $\sin(\theta)$ centered at $a = 10^\circ$.

```
def sin_taylor(theta, a, N): """Approximate sin(theta) using Taylor centered at a"""
    result = 0.0
    sin_a, cos_a = math.sin(a), math.cos(a)
    for n in range(N): # Derivatives of sin cycle every 4 terms
        derivative_pattern = n % 4
        if derivative_pattern == 0: f_deriv = sin_a
        elif derivative_pattern == 1: f_deriv = cos_a
        elif derivative_pattern == 2: f_deriv = -sin_a
        else: f_deriv = -cos_a
        term = f_deriv * ((theta - a) ** n) / math.factorial(n)
        result += term
    return result
```

Comparison: For the same angles (1° – 30°), calculate errors using:

- Maclaurin (centered at 0°)
- Taylor (centered at 10°)

Which performs better near 10° ? Which is better far from 10° ?

The Taylor series centered at 10° performs better near 10° . The Maclaurin series centered at 0° is better farther from 10° .

PART 6: ENGINEERING DECISION—ERROR TOLERANCE

Requirement: The approximation must have less than 0.1% error.

Task 1: For each angle (1° – 30°), find the minimum number of terms needed to meet the 0.1% tolerance using both Maclaurin and Taylor approximations.

Task 2: Investigate the small-angle approximation $\sin(\theta) \approx \theta$. At what angle does this 1-term approximation stop being sufficiently accurate?

Analysis:

1. Does adding more terms always improve the approximation?

Yes, adding more terms usually makes the approximation more accurate.

2. Why is the Maclaurin approximation most accurate near zero?

Because the Maclaurin series is centered at zero.

3. Why can changing the Taylor expansion point improve accuracy?

It can improve accuracy because the center is closer to the value I want to find.

4. How does the distance from the expansion point affect accuracy?

The farther the value is from the center, the less accurate the approximation can be.

5. What is the critical angle for the $\sin(\theta) \approx \theta$ approximation?

Based on my results, the critical angle is 5° , where the error goes above 0.1%.

PART 7: FINAL ENGINEERING RECOMMENDATION

Decision Problem:

You are the civil engineer responsible for the calculation. You need an approximation with less than 0.1% error. Would you use the Maclaurin series, a Taylor series centered closer to the angle of interest, or the exact sine function? Defend your choice using your numerical results.

Your recommendation must include:

- Number of terms required
- Percentage error achieved
- Convergence behavior
- Computational simplicity vs. accuracy tradeoff
- Valid angle range for your solution

I would use the Maclaurin series with 2 terms because it is simple and accurate enough for this problem.

- **Number of terms: 2 terms**
- **Percentage error: Less than 0.1%; at 30°, the error is about 0.065%**
- **Convergence: Adding more terms makes the answer closer to the exact value.**
- **Simplicity vs. accuracy: 2 terms are easy to calculate while still giving enough accuracy.**
- **Valid angle range: 1°–30° based on my results.**

REQUIRED PYTHON OUTPUT

Your final Python program must produce the following deliverables:

1. Numerical Tables

Formatted tables showing exact values, approximations, and errors for each angle and number of terms.

2. Convergence Plot

A figure showing how percentage error decreases as you add more terms, for different angles.

3. Function Comparison Plot

Side-by-side plots of exact $\sin(\theta)$ vs. Maclaurin and Taylor approximations with different numbers of terms.

4. Error Comparison Plot

Visual comparison of absolute and percentage errors between Maclaurin and Taylor series.

5. Written Recommendation

A clear explanation of your engineering decision with supporting numerical evidence.

Implementation Note: Explicitly implement the series summation using Python loops. Do not hide the mathematics inside symbolic mathematics software. The purpose is to make you see the progression from geometric series → power series → Maclaurin → Taylor → engineering approximation → error analysis → engineering decision.

THE MATHEMATICAL PROGRESSION

1. Geometric Series

$$1 + x + x^2 + \dots = 1/(1-x)$$



2. Power Series

$$a_0 + a_1 x + a_2 x^2 + \dots$$



3. Maclaurin Series

Expand around $\theta = 0$



4. Taylor Series

Expand around any point $\theta = a$



5. Engineering Approximation

Apply to $y = L \sin(\theta)$



6. Error Analysis

Compare with exact values



7. Engineering Decision

What's the best tradeoff?